

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
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S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 903/2
Bridge Chainage – 930+450

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MJB

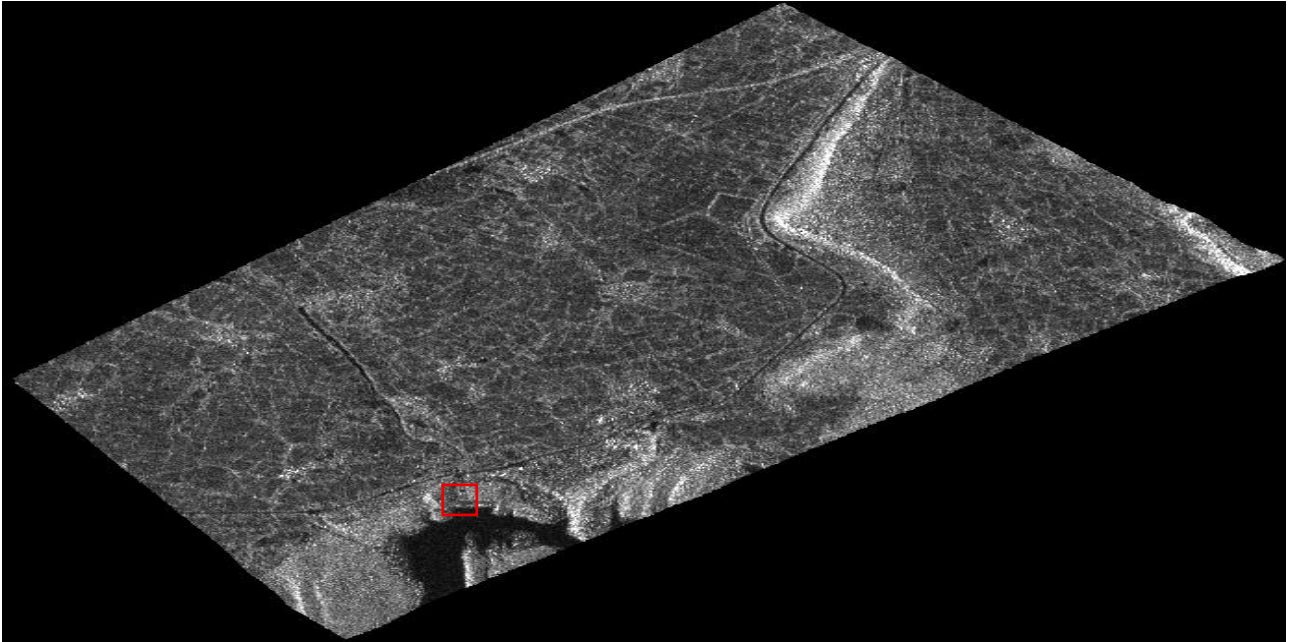
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

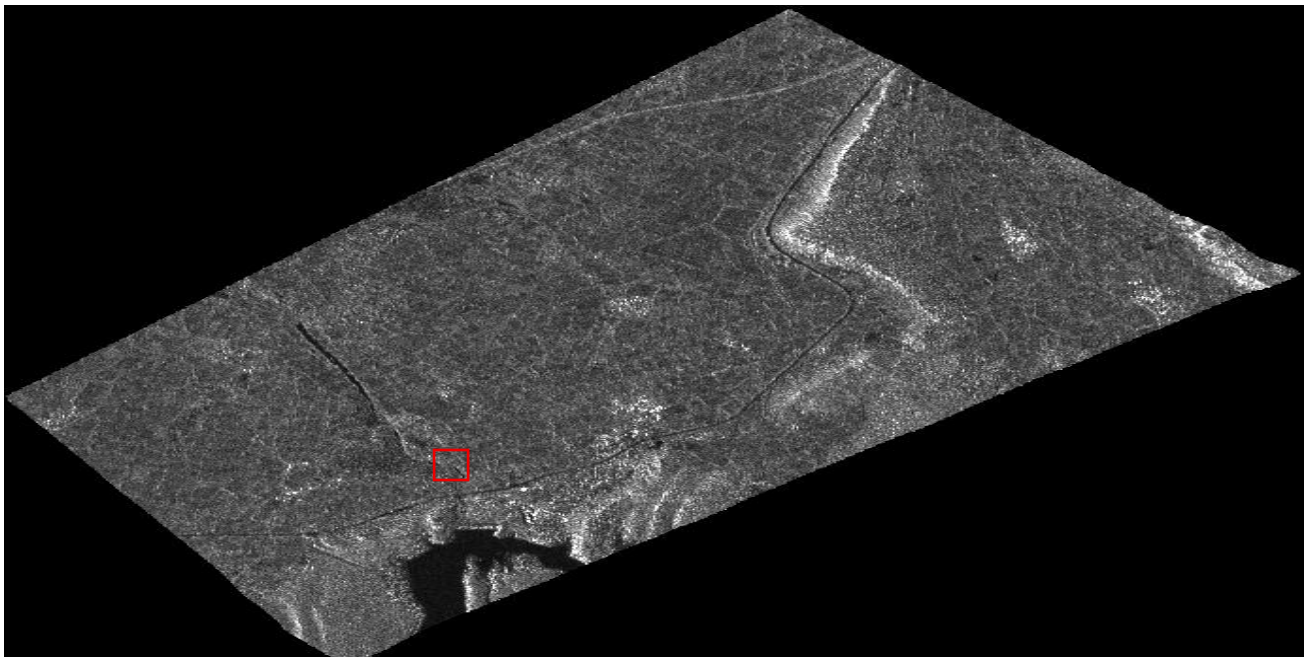
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

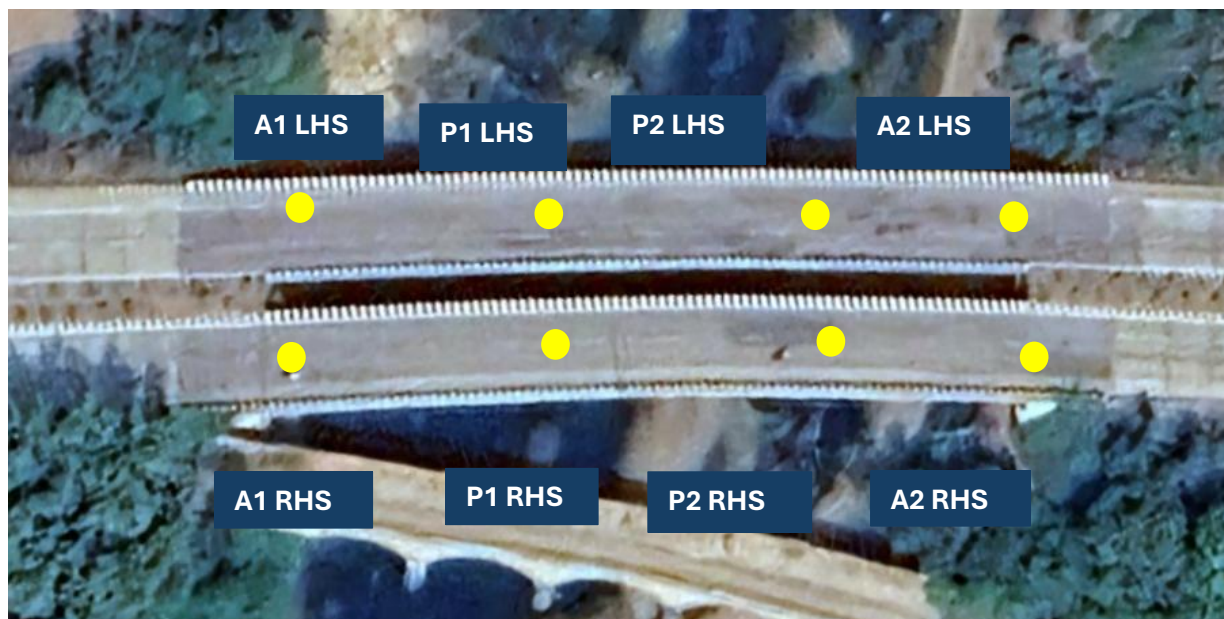


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MJB – CH-938+250

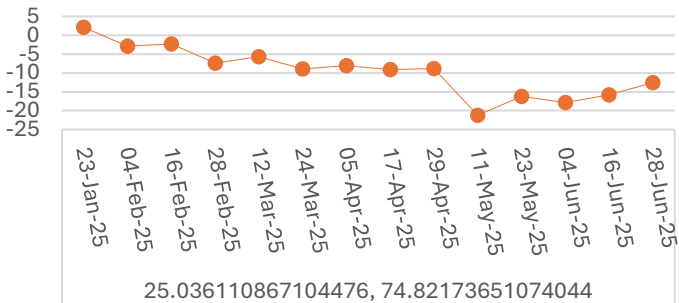
Length – 56.25M



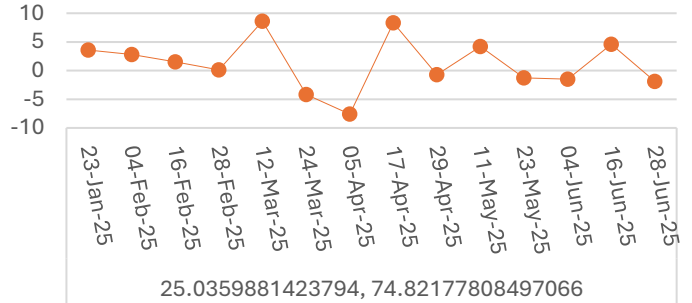
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
P2		Pier 2	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 to 28 June 2025

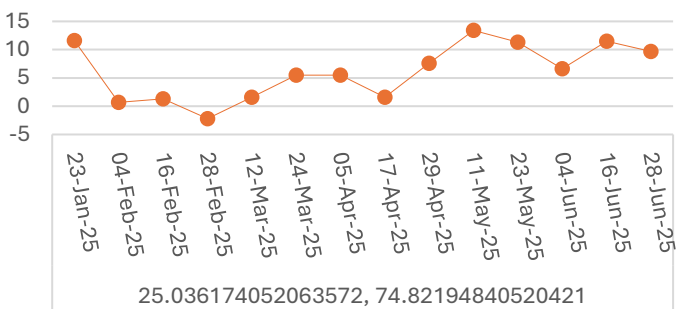
Displacement @A1 LHS VV



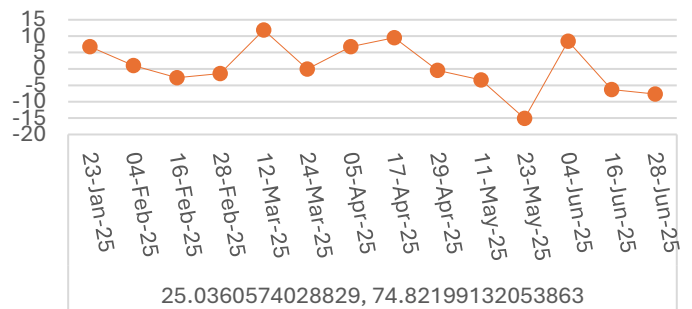
Displacement @A1 RHS VV



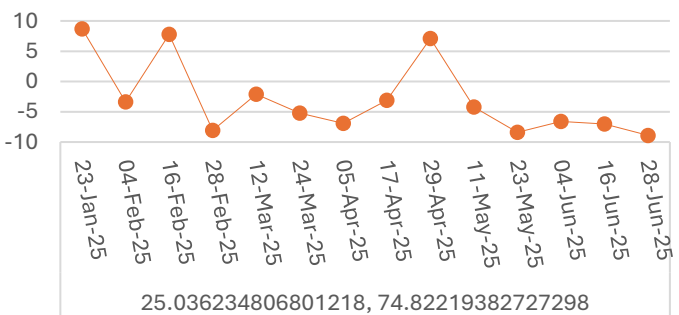
Displacement @P1 LHS VV



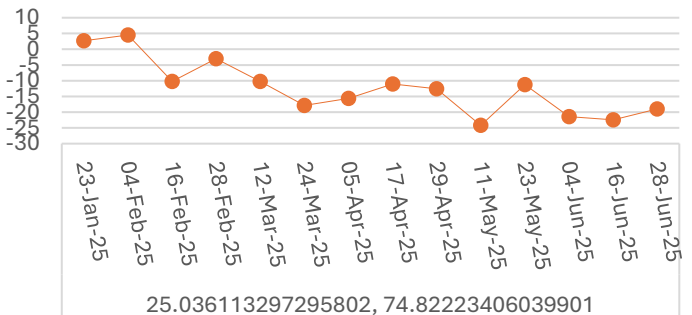
Displacement @P1 RHS VV



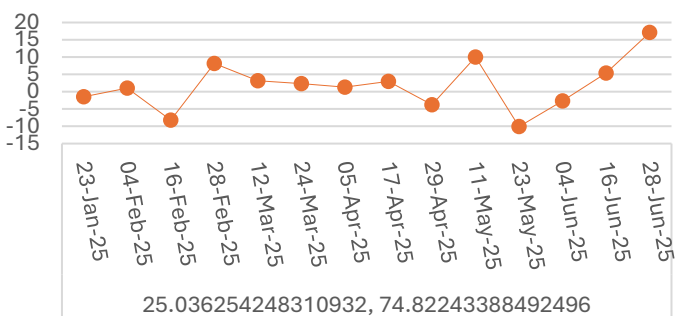
Displacement @P2 LHS VV



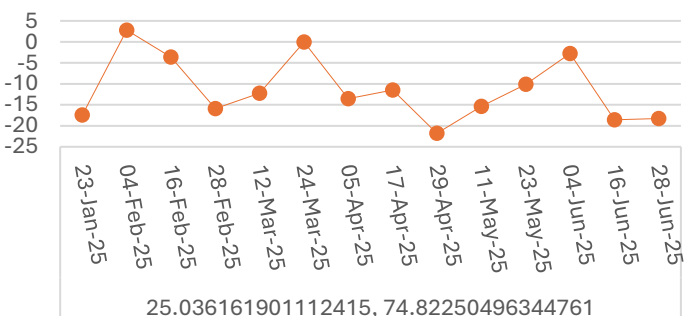
Displacement @P2 RHS VV



Displacement @A2 LHS VV

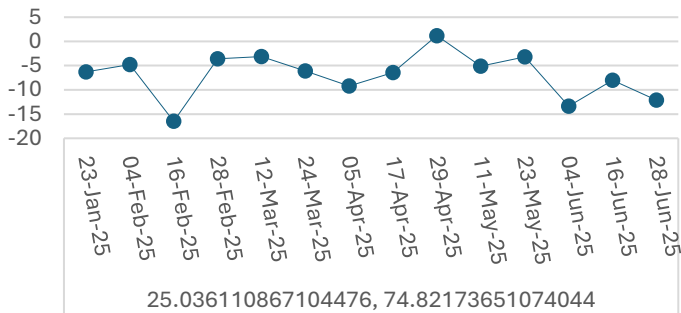


Displacement @A2 RHS VV

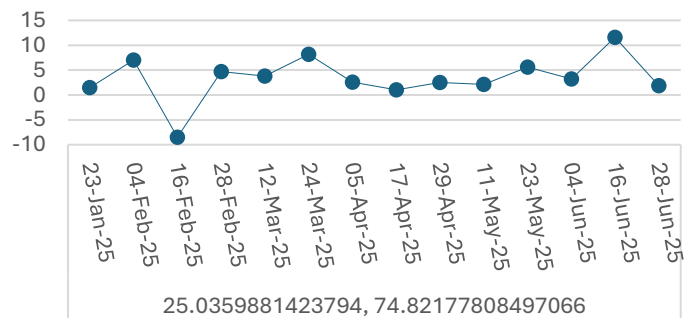


3.2 VH Displacement From 23 Jan 2025 to 28 June 2025

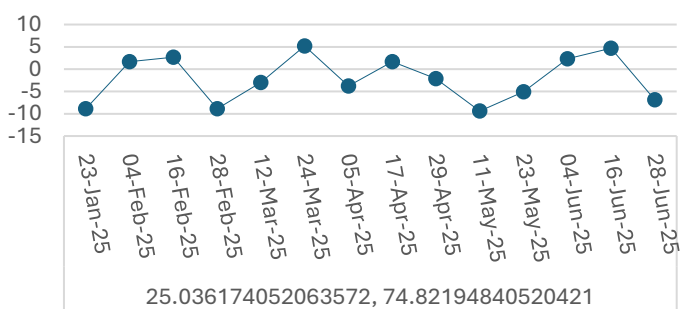
Displacement @A1 LHS VH



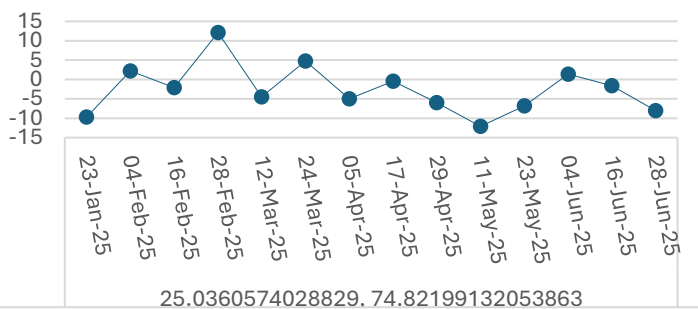
Displacement @A1 RHS VH



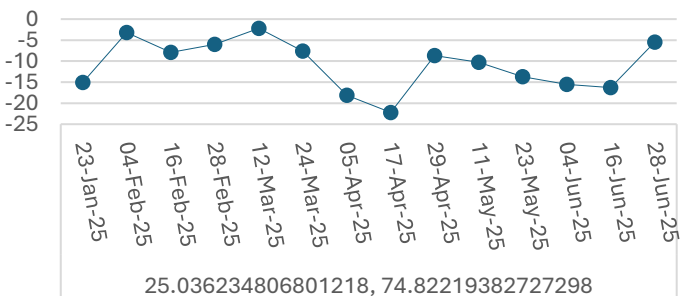
Displacement @P1 LHS VH



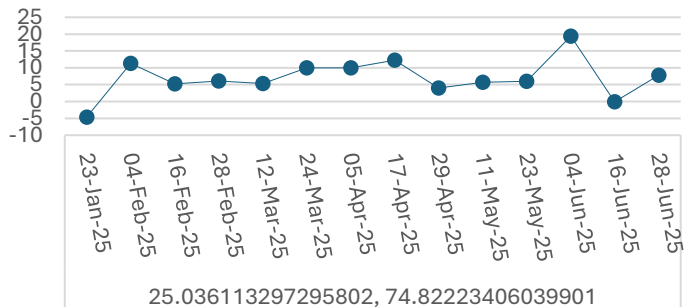
Displacement @P1 RHS VH



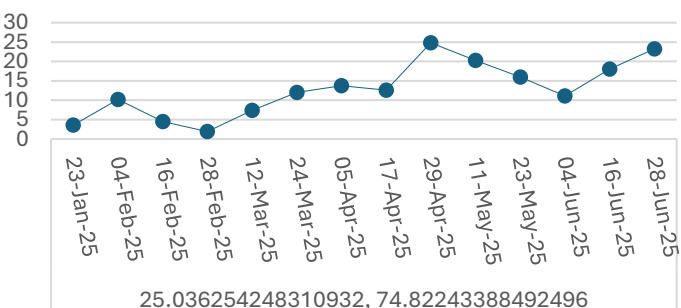
Displacement @P2 LHS VH



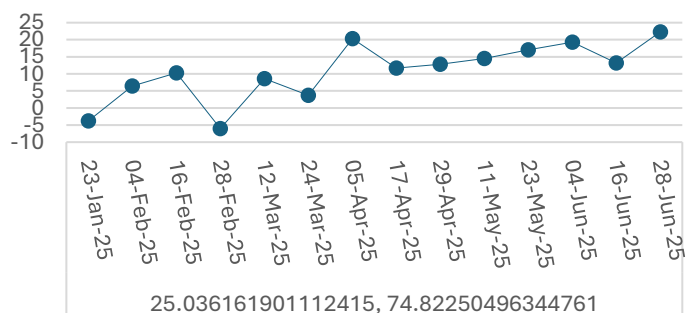
Displacement @P2 RHS VH



Displacement @A2 LHS VH



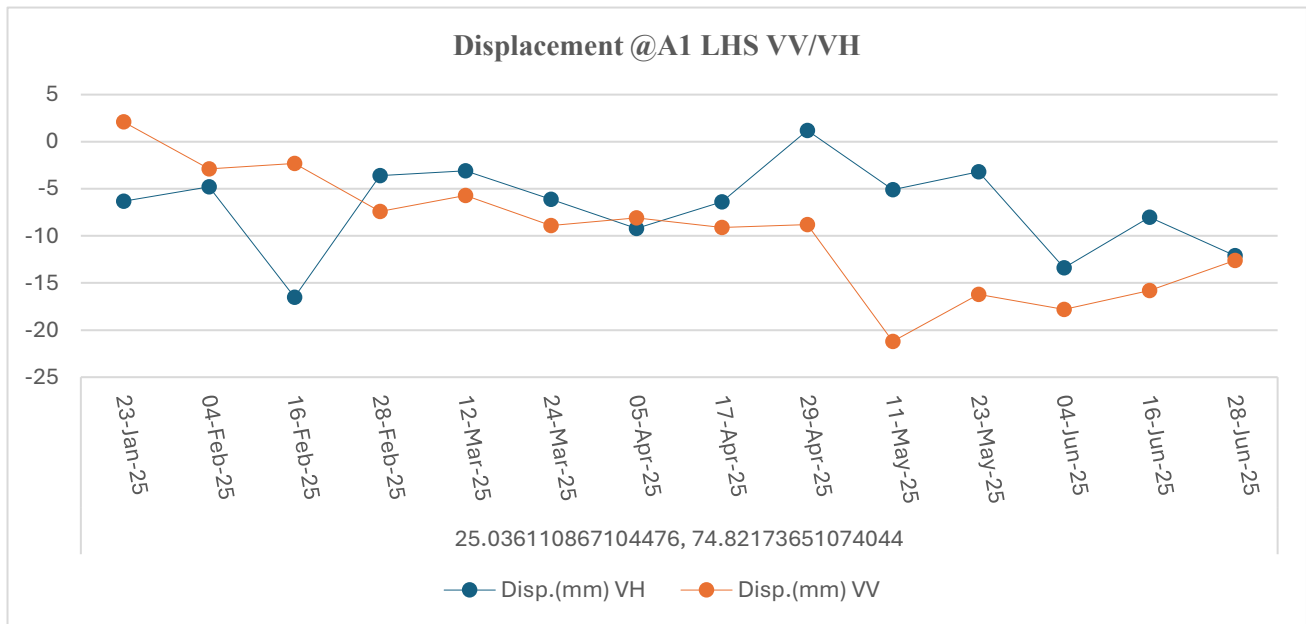
Displacement @A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.036110867104476, 74.82173651074044)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains predominantly negative over the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the A1 LHS location. VV values range approximately from +2 mm to –22 mm.
- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between +1 mm and –17 mm. VH alternates between short-lived recoveries and negative excursions, but remains largely settlement-dominated, suggesting persistent lateral movement in a preferred direction.

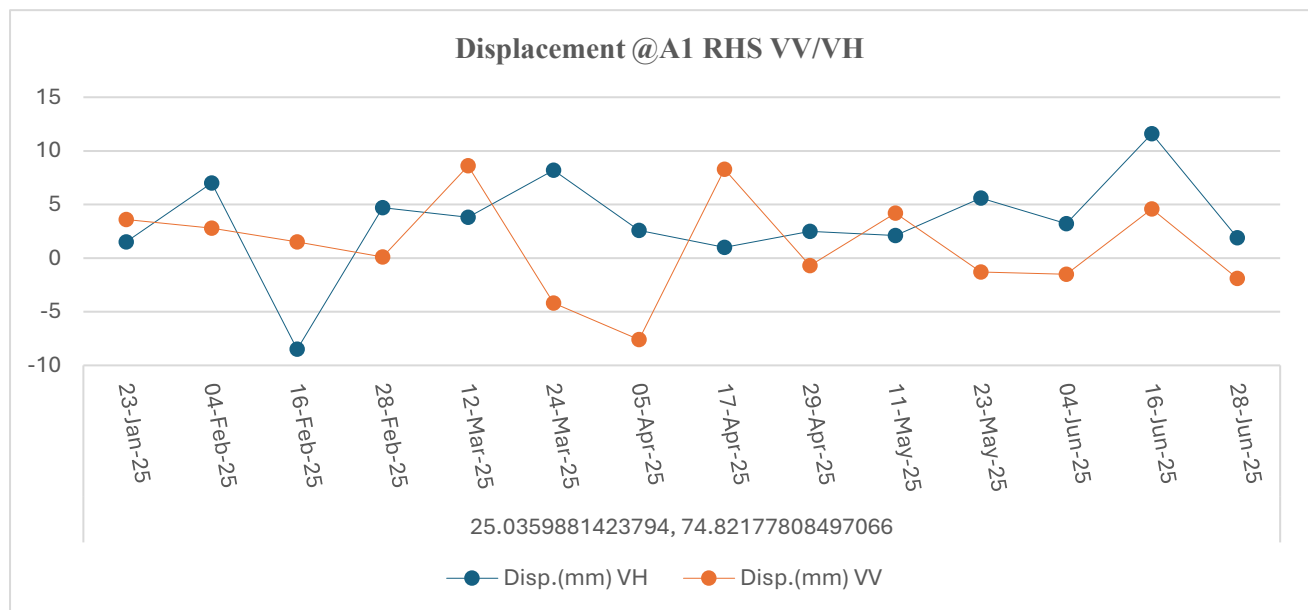
Critical Events & Anomalies:

- **March 2025:** VH reaches a local positive peak (+1 mm), while VV remains negative (–9 mm), highlighting a brief lateral rebound without vertical recovery.
- **June 2025:** VH recovers slightly toward –8 mm, while VV improves marginally to –12 mm, suggesting post-event adjustment rather than true recovery.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.0359881423794, 74.82177808497066)



Key Observations:

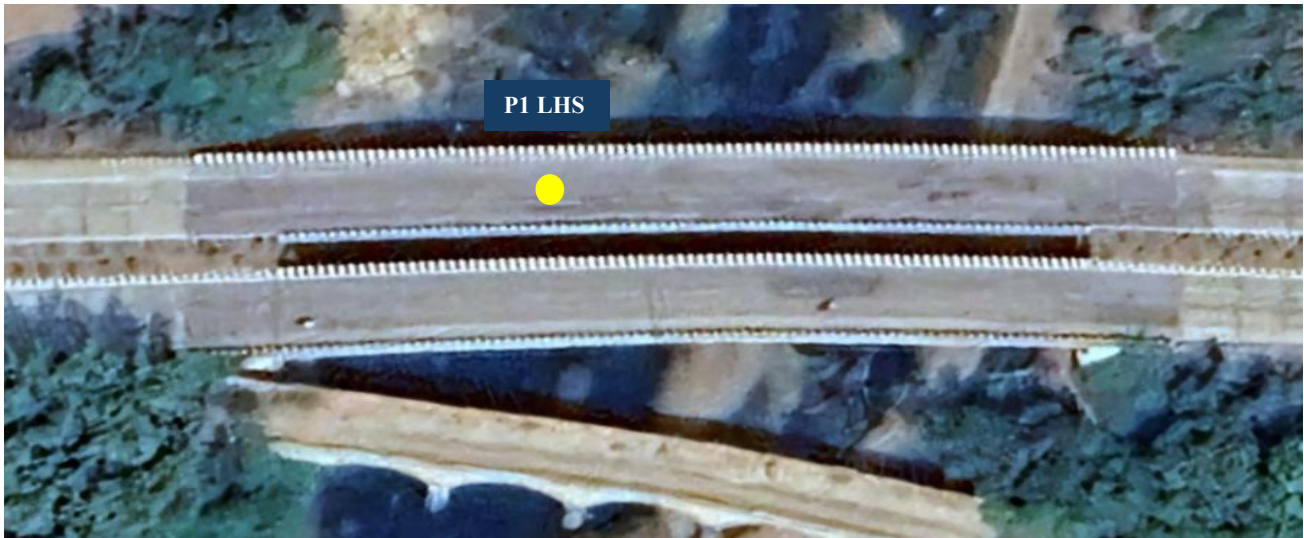
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between +12 mm and –9 mm.
- **Vertical displacement (VV)** Vertical displacement shows moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from +8 mm to –8 mm.

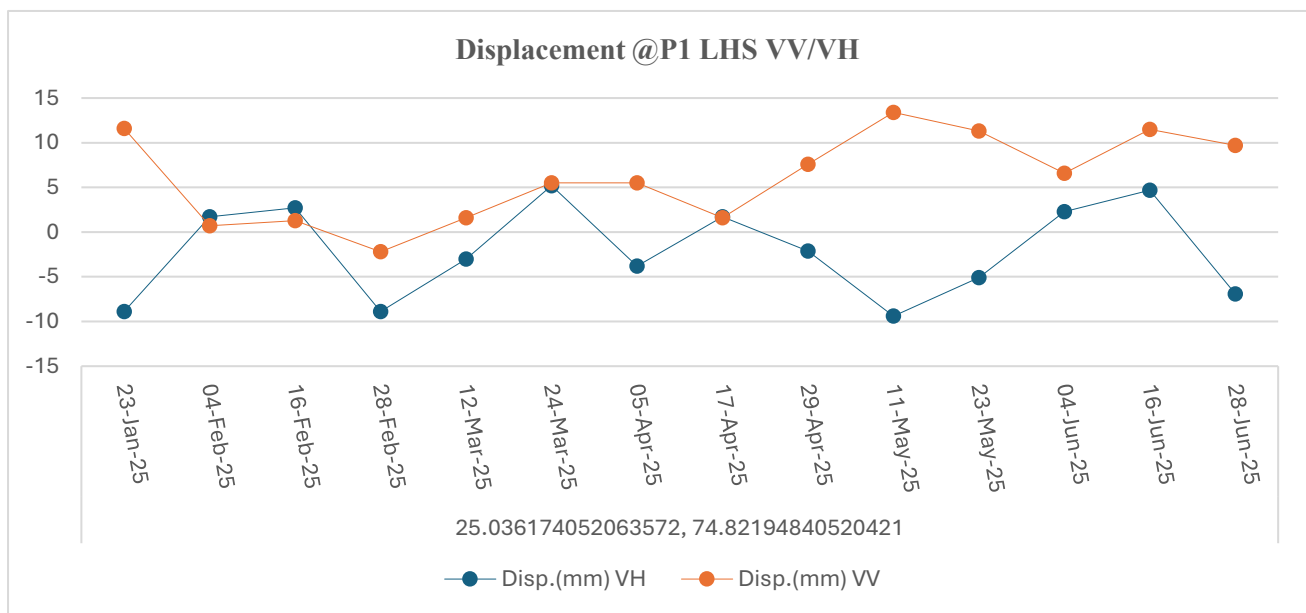
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+7 mm), while VV remains mildly positive (+2 mm), indicating a short-lived lateral uplift event with limited vertical response.
- **June 2025:** VH declines to +2 mm, while VV drops to –2 mm, confirming continued oscillatory behaviour with no clear stabilization toward the end of the monitoring window

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.036174052063572, 74.82194840520421)



Key Observations:

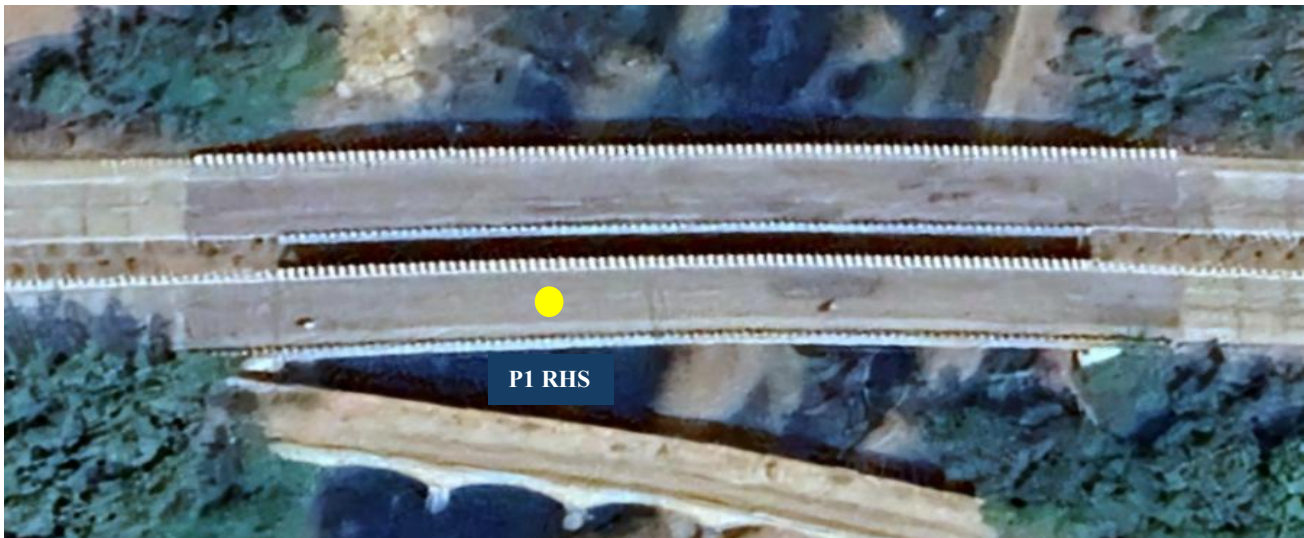
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains predominantly positive throughout the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the P1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between +5 mm and –10 mm.

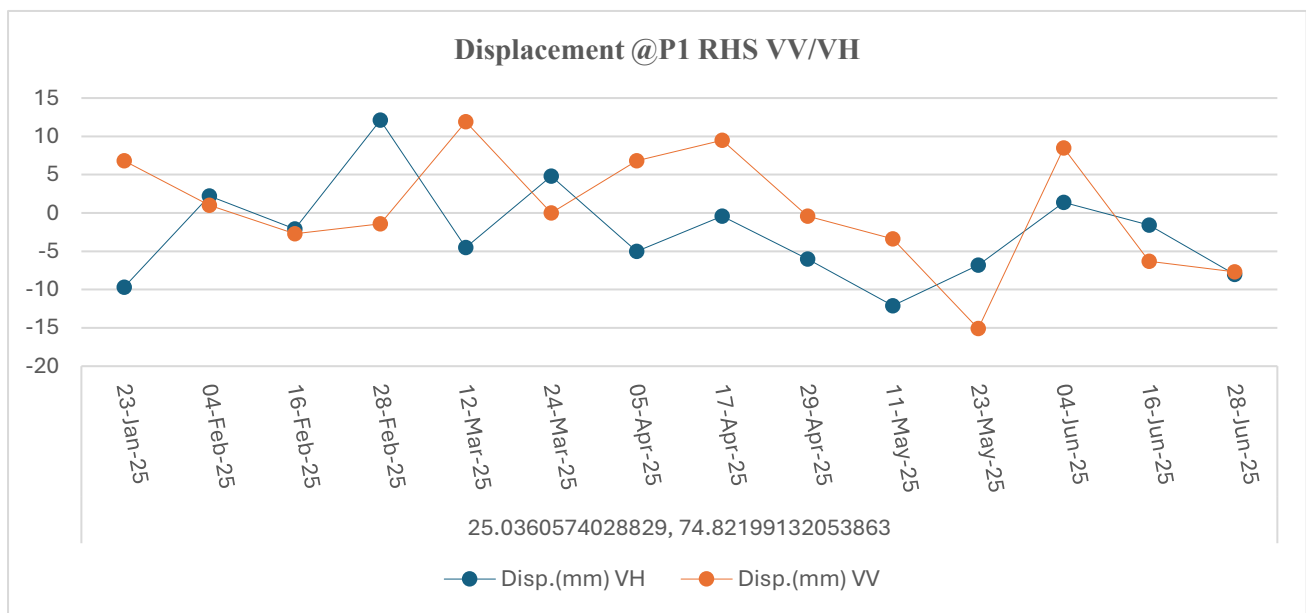
Critical Events & Anomalies:

- **February 2025:** VH recovers to +1–2 mm, while VV remains positive (+1 mm), suggesting a short-lived lateral stabilization under continued uplift.
- **May 2025:** A distinct divergence occurs—VV rises sharply to around +12 mm, while VH simultaneously drops to about –10 mm, indicating conflicting vertical uplift and lateral movement at the same time
- **June 2025:** VH drops again to –7 mm, while VV remains positive (+9 mm), confirming continued lateral instability despite stable vertical uplift.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.0360574028829, 74.82199132053863)



Key Observations:

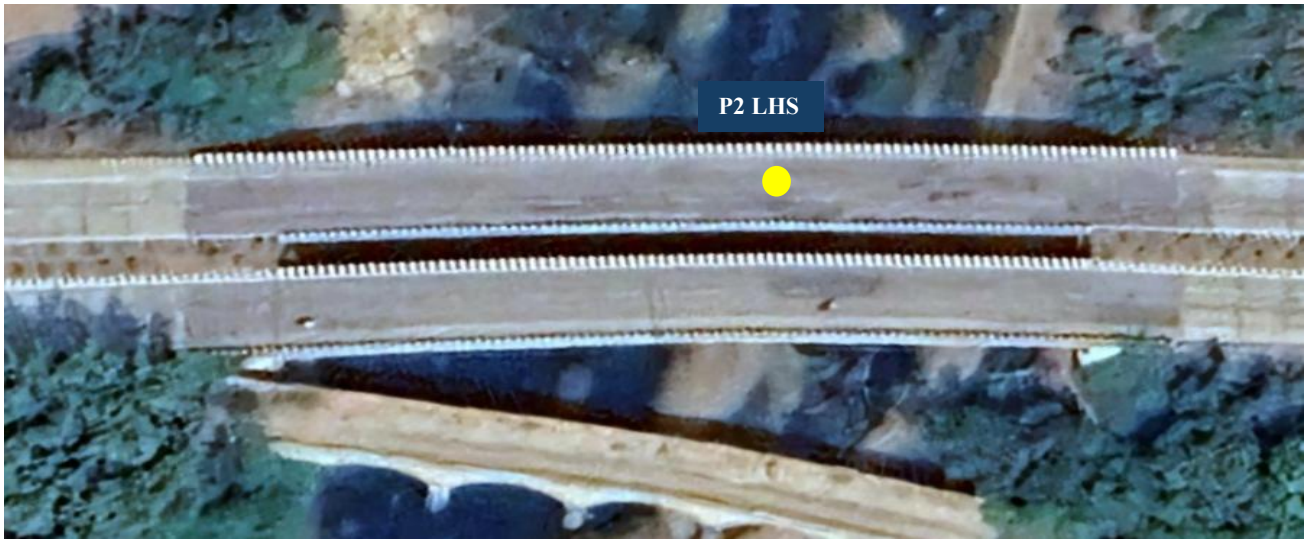
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits high variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from +12 mm to –14 mm.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate to high variability, ranging roughly between +12 mm and –12 mm.

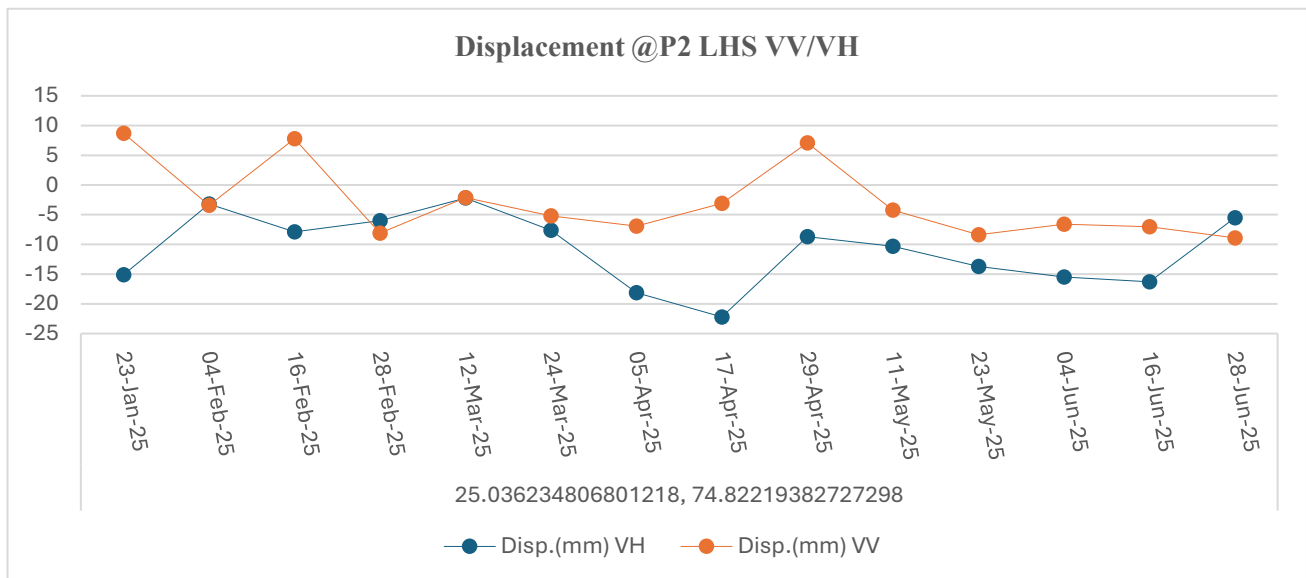
Critical Events & Anomalies:

- **January 2025:** VV starts at a moderate uplift level (+7 mm), while VH is negative (–9 mm), indicating initial vertical uplift accompanied by lateral settlement.
- **May 2025:** VH rebounds to +1 mm, while VV rises sharply to +8 mm, representing a short-lived coupled uplift phase.
- **June 2025:** Both VH and VV settle near negative values (–7 mm), indicating no clear stabilization trend by the end of the monitoring period.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (25.036234806801218, 74.82219382727298)



Key Observations:

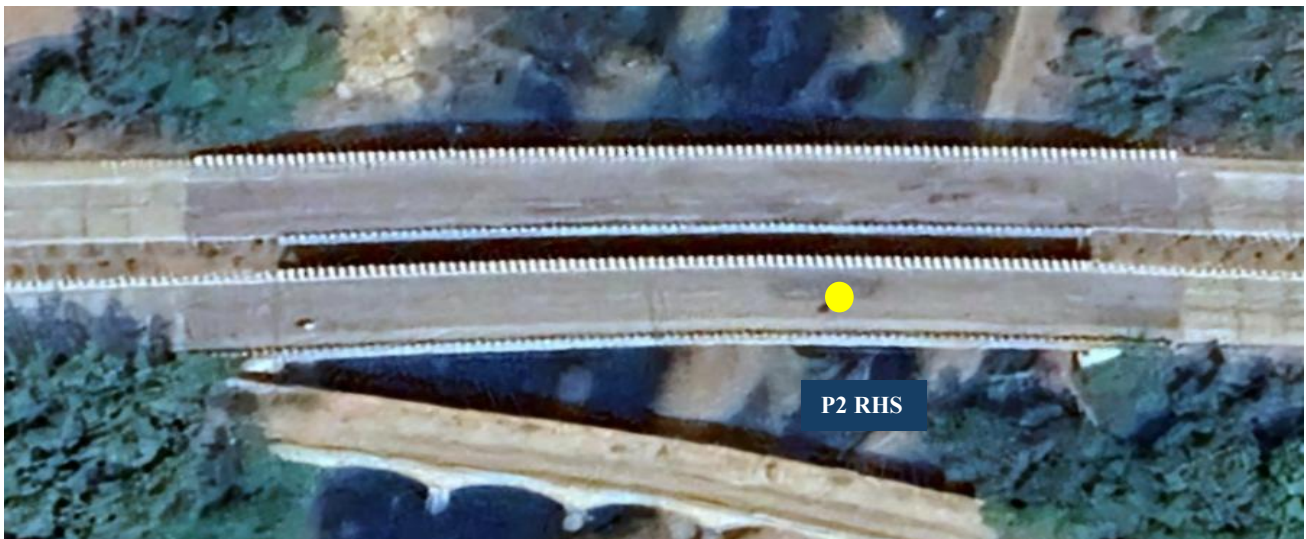
Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement exhibits strong negative dominance with high variability, ranging roughly between -7 mm and -22 mm. VH remains consistently negative for most of the monitoring period, indicating persistent lateral settlement with only brief, low-magnitude recoveries.
- **Vertical Displacement (VV)** Vertical displacement shows moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from $+8$ mm to -9 mm.

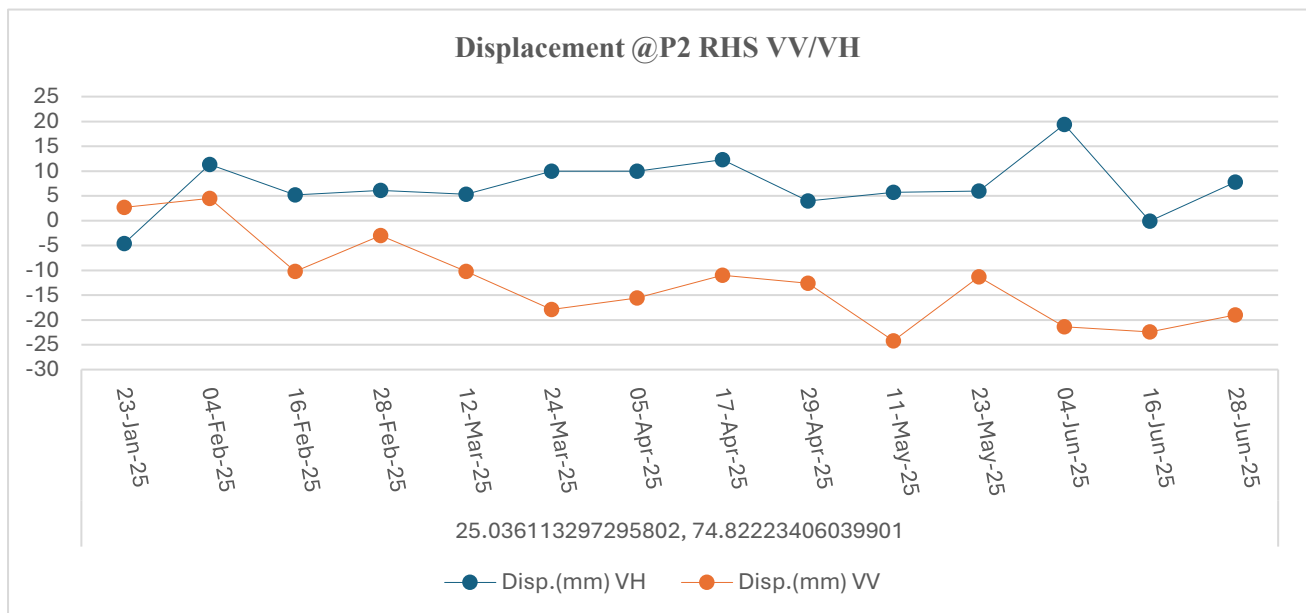
Critical Events & Anomalies:

- **May 2025:** VH declines again to -12 to -14 mm, while VV drops to -9 mm, reflecting renewed coupled settlement behaviour.
- **June 2025:** VH remains negative (-15 to -8 mm), while VV stabilizes around -8 to -9 mm, confirming continued deformation with no clear recovery trend toward the end of the monitoring period.

3.8 SAR Based Analysis at Pier 2 RHS:



P2 RHS (25.036113297295802, 74.82223406039901)



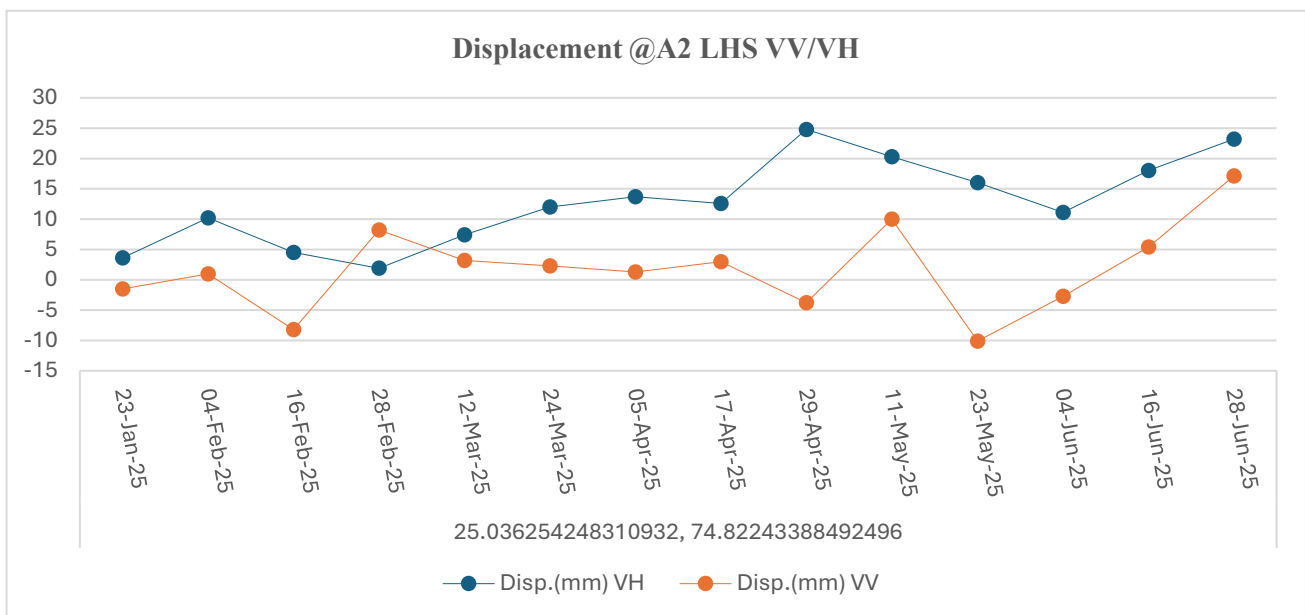
Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period (Jan–Jun 2025), indicating persistent vertical settlement at the P2 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability with predominantly positive values, ranging roughly between +2 mm and +20 mm.

Critical Events & Anomalies:

- **January 2025:** VH starts slightly negative (–4 mm), while VV is near 0 mm, indicating initial lateral adjustment with minimal vertical settlement.
- **April 2025:** VH remains stable, while VV fluctuates between –12 and –18 mm, indicating persistent settlement with limited vertical recovery
- **June 2025:** VH drops briefly to +2 mm before recovering to +8 mm, while VV remains negative confirming continued deformation with no stabilization toward the end of the monitoring period.



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025).
- **Horizontal Displacement (VH)** Horizontal displacement exhibits a strong and persistent positive trend, ranging roughly between +3 mm and +25 mm.

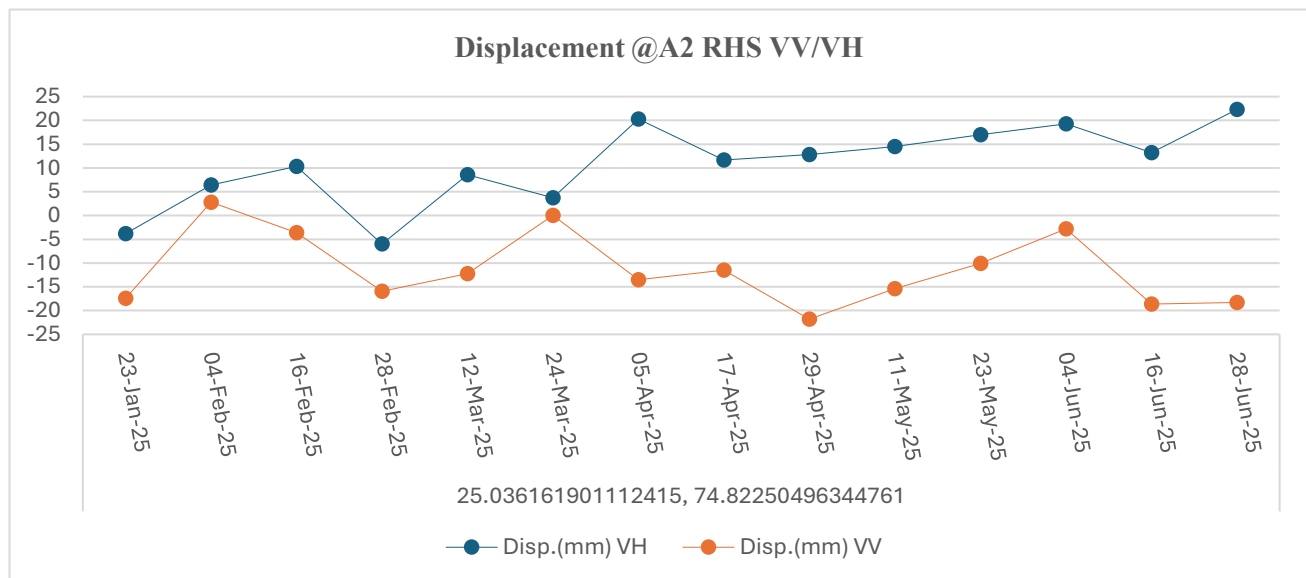
Critical Events & Anomalies:

- **Feb 2025:** VV drops to –8 mm, while VH remains positive (+5 mm), marking a decoupled deformation event with vertical settlement occurring under continued lateral movement
- **June 2025:** Both VH and VV increase toward the end of the period (VH +18 to +23 mm, VV +5 to +17 mm), representing a combined lateral displacement and vertical uplift intensification.

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.036161901112415, 74.82250496344761)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains predominantly negative throughout the monitoring period (Jan–Jun 2025), indicating persistent vertical settlement at the A2 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits a strong and progressively increasing positive trend, ranging roughly between -4 mm and $+23$ mm.

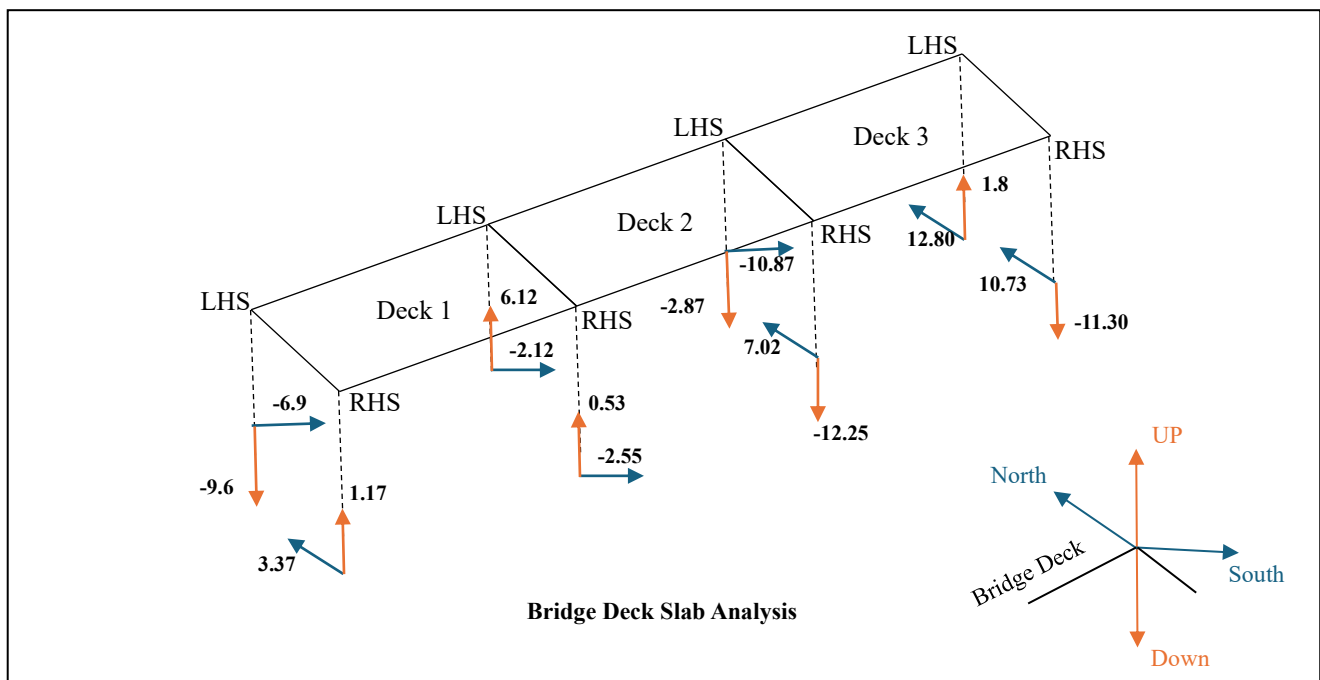
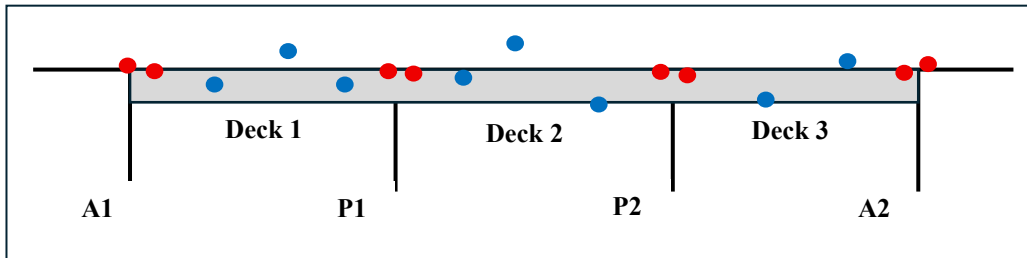
Critical Events & Anomalies:

- **January 2025:** VH begins slightly negative (-4 mm), while VV is already strongly negative (-18 mm), indicating early-stage combined lateral and vertical settlement.
- **May 2025:** VH continues increasing ($+15$ to $+18$ mm), while VV partially recovers to -10 mm, reflecting continued lateral acceleration with limited vertical relief.
- **June 2025:** VV improves briefly to -3 mm, while VH reaches $+20$ mm, indicating a temporary vertical adjustment under strong lateral displacement.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits distinctly non-uniform vertical displacement along its longitudinal direction, indicating differential vertical response across Decks 1 to 3. The maximum downward settlement is observed at Deck 2 (RHS), reaching approximately -12.25 mm, identifying this region as the most vertically critical location.

Horizontal Displacement Behaviour:

- Horizontal displacement shows strong directional variability across the deck width and along its length, indicating a torsion-dominated deformation response. The maximum horizontal displacement is recorded at Deck 3 (LHS), reaching approximately +12.80 mm, representing dominant lateral movement at the terminal deck region.

CONCLUSION:

- The bridge deck slab demonstrates a complex deformation pattern, characterized by simultaneous vertical settlement, localized uplift, and asymmetric horizontal displacement across successive decks. The concentration of maximum settlement at Deck 2, combined with increasing horizontal displacement toward Deck 3, suggests progressive torsional influence along the deck length.